Sam: $\forall \varepsilon > 0, \varepsilon \in \mathbb{R}, \exists N_1 \in \mathbb{N}$, s.t. $\forall n \geq N_1, n \in \mathbb{N}, \boxed{|y_n - a| < \varepsilon}$

$\therefore \boxed{a - \varepsilon < y_n < a + \varepsilon} \Rightarrow \boxed{0 - \varepsilon < (y_n - a) < 0 + \varepsilon}$

$\Rightarrow \boxed{|(y_n - a) - 0| < \varepsilon}$. $\boxed{\therefore \lim_{n \to \infty} d_n = \lim_{n \to \infty} (y_n - a) = 0}$ $\square$

Author: Now I invite you to prove the following

Theorem: If (y_n) is a bounded sequence and (d_n) is infinitesimal, then $(y_n d_n)$ is infinitesimal as well.

Sam: **Proof:** Since y_n is a bounded sequence, by defn,

$\exists A, B \in \mathbb{R}$ s.t. $\forall n \in \mathbb{N}, \boxed{A \leq y_n \leq B}$ ①

$\because d_n$ is an infinitesimal sequence, by defn its limit is 0.

$\boxed{d_n = x_n - a}$, i.e. there exists a convergent sequence (x_n) whose limit is a.

$\therefore \forall \varepsilon > 0, \varepsilon \in \mathbb{R}, \exists N_1 \in \mathbb{N}$, s.t. $\forall n \in \mathbb{N}, n \geq N_1, |d_n - 0| < \varepsilon$

$\Rightarrow 0 - \varepsilon < d_n < 0 + \varepsilon \Rightarrow \boxed{-\varepsilon < d_n < \varepsilon}$ ②

Multiplying ① and ②, we get: $\boxed{-A\varepsilon < y_n d_n < B\varepsilon}$

or, $\boxed{0 - A\varepsilon < y_n d_n < 0 + B\varepsilon}$. Let $C = \max(A, B)$

or, $\boxed{0 - C\varepsilon < y_n d_n < 0 + C\varepsilon} \Rightarrow \boxed{0 - \varepsilon_1 < y_n d_n < 0 + \varepsilon_1 \text{ where } \varepsilon_1 = C\varepsilon}$

$\Rightarrow |y_n d_n - 0| < \varepsilon_1$ $\boxed{\therefore \lim_{n \to \infty} y_n d_n = 0}$ $\square$

(Does it really hold for $\forall \varepsilon_1$) ?

Reader: Let us select an arbitrary $\varepsilon > 0$. We must prove that there is N such that for all $n \geq N$ the terms of the sequence $(y_n d_n)$ satisfy the inequality $|y_n d_n| < \varepsilon$.

Author: Do you mind a hint? As the sequence (y_n) is bounded, one can find M such that $|y_n| \leq M$ for any n.